

SAQ for Compact, Calibrated ECG Transformers

A core-AI framing: efficient representation learning for structured human physiological signals

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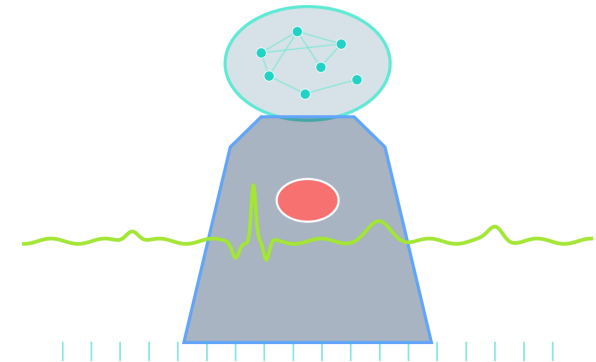
Poster abstract

Human biosignals such as ECGs are long, noisy and physiologically structured. This work studies Segment-Aware Quantisation (SAQ), a physiology-guided tokenisation scheme that converts raw 12-lead ECG waveforms into compact transformer-ready sequences. SAQ allocates representation precision by cardiac segment importance - highest for ST, lower for PR - then feeds a lightweight transformer with post-hoc calibration and clinically constrained thresholds. The result is not a generic ECG wrapper: it is an efficient representation-learning pipeline for human-centred AI, with a clear path to scalable HPC experimentation and low-resource inference.

Key idea: compress ECG by physiological morphology, not blind patching.

Visual thesis

Core AI for structured physiological sequences



Human physiology + structured sequence AI + compact tokenisation

0.959

Macro-AUROC

0.921

Macro-AUPRC

6.9M

parameters

47 ms

T4 GPU latency

0.127

post-calibration ECE

Retrospective experimental findings only. Prospective clinical evaluation is required before any patient-care use.

2 Problem: Long, Structured Human Signals

Why this is core AI, not a cosmetic healthcare app

Representation

Raw 12-lead ECGs contain tens of thousands of samples per 10-second record. The modelling problem is long-sequence representation learning.

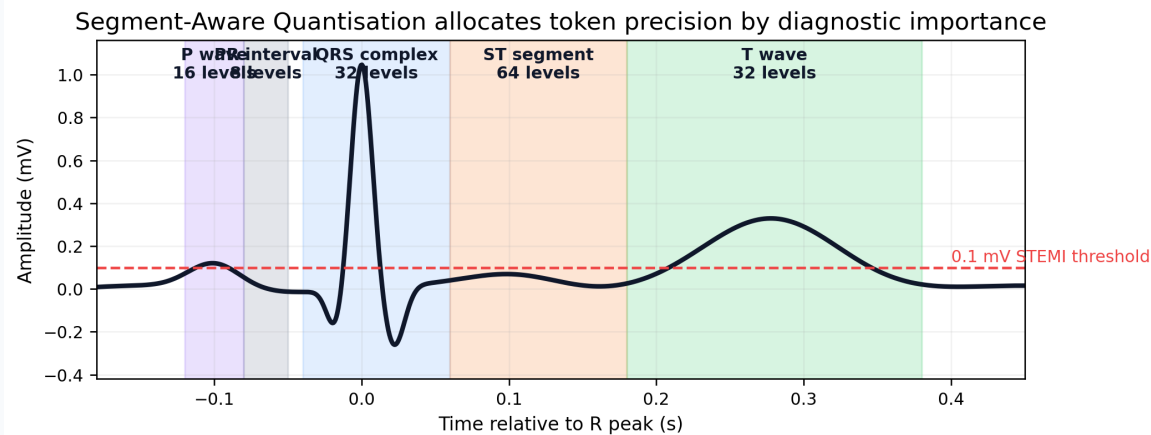
Inductive bias

Blind patching or generic compression can erase small but clinically critical morphology, especially ST-segment changes.

Trust + compute

A useful model needs discrimination, calibration, threshold control and low latency - not just a high internal accuracy number.

Physiology-aware token budget



SAQ assigns more precision to diagnostic segments and less to low-information regions.

SAQ allocation

ST segment	64 levels
QRS complex	32 levels
T wave	32 levels
P wave	16 levels
PR interval	8 levels

~512

tokens after downstriding

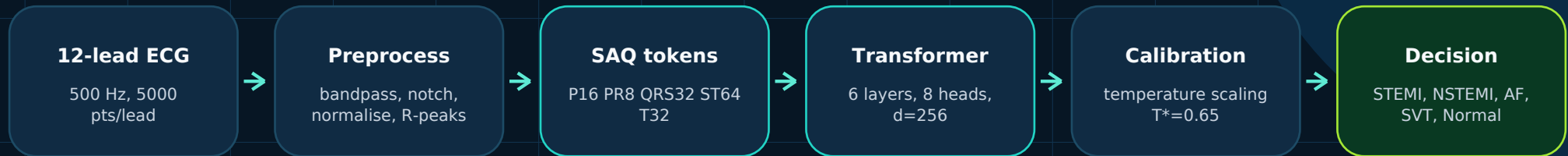
58-75%

sequence reduction range

0.082 mV

ST RMSE fidelity

Design claim: SAQ preserves diagnosis-critical morphology while cutting sequence length and compute pressure.



Model architecture

- Segment-specific embeddings preserve cardiac-region identity.
- Cycle-aware positional encoding anchors sequence position to heartbeat periodicity.
- 6-layer transformer with 8 attention heads, FFN=1024.
- Approx. 6.9M parameters, 26-28 MB fp32 footprint.

Training regime

- Phase 1: warm-up with cross-entropy + morphology fidelity loss.
- Phase 2: class-balanced focal loss with inverse-frequency priors.
- Phase 3: hard-negative mining for STEMI/NSTEMI and NSTEMI/AF confusions.
- Validation-only calibration and threshold selection.

Clinical operating points

Class	Policy	Threshold	Sensitivity
STEMI	Sens $\geq$.95	.243	.952
NSTEMI	Max F2	.341	.864
AF	Youden J	.398	.891
SVT	Youden J	.412	.875
Normal	Youden J	.427	.903

What makes this research-oriented?

- Inductive bias: domain structure is encoded in the representation before the model sees the sequence.
- Calibration: the pipeline treats probability quality as a first-class objective, not an afterthought.
- Compute efficiency: compression reduces sequence burden and supports scalable experimentation on GPU/HPC resources.

0.959

Macro-AUROC

PTB-XL test

0.921

Macro-AUPRC

PTB-XL test

95.2%

STEMI sensitivity

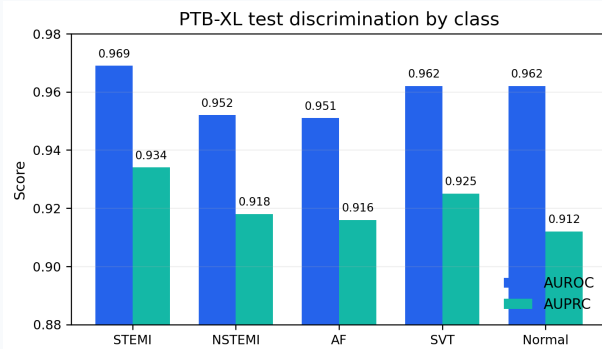
with constrained threshold

0.851

CPSC AUROC

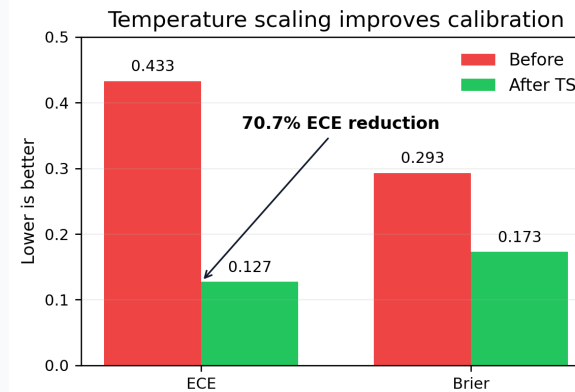
frozen transfer

Per-class discrimination



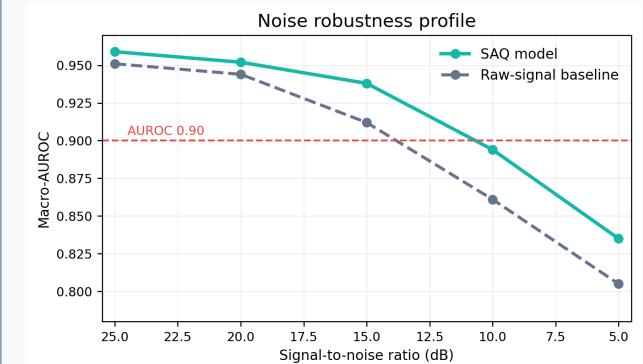
PTB-XL test set.

Calibration



Temperature scaling reduces ECE.

Noise robustness



AUROC remains above 0.90 around 15 dB SNR.

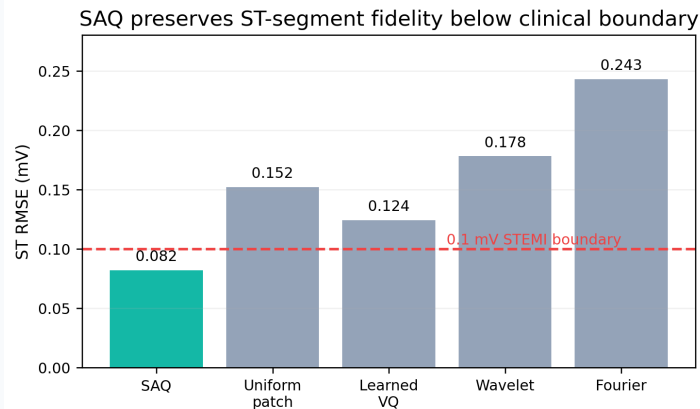
Calibration + threshold outcome

- ECE: 0.433 -> 0.127 after temperature scaling.
- Brier: 0.293 -> 0.173.
- STEMI: 95.2% sensitivity at 88.3% specificity; NPV 0.976 at dataset prevalence.

Interpretation

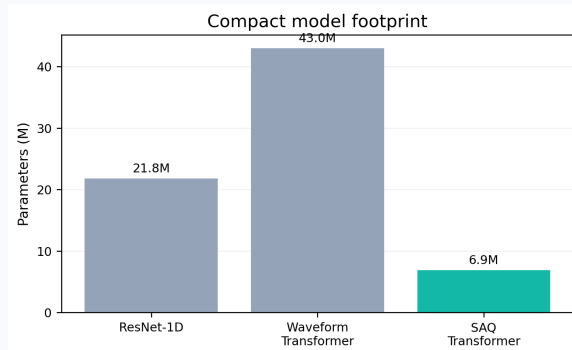
The strongest evidence is not only the AUROC. The stronger story is the full stack: structured tokenisation, compact model size, calibration, controlled thresholding, cross-dataset transfer and noise stress-testing. That is what makes the work defensible in an HPC research environment.

Compression fidelity



Only SAQ preserves ST RMSE below the 0.1 mV boundary in the matched comparison.

Efficiency comparison



Proposed model is smaller than internal baselines trained under the same protocol.

Why this belongs at IHPCCS

- Long-sequence AI: ECG modelling is a sequence compression and representation-learning problem.
- Performance engineering: token length, memory footprint and latency directly affect deployment feasibility.
- Scalable validation: larger cohorts require reproducible pipelines, distributed sweeps and robust external testing.
- HPC bridge: the next step is profiling, workflow automation and multi-dataset experimentation on HPC resources.

Next technical steps

GPU profiling

Benchmark kernels, memory use and end-to-end latency across T4/V100/HPC nodes.

Distributed HPO

Run controlled hyperparameter and ablation sweeps with reproducible manifests.

External validation

Scale to broader ECG datasets, acquisition devices and patient subgroups.

Workflow packaging

Containerise training, calibration and evaluation for repeatable execution.

Limitations kept explicit

Retrospective design; custom five-class taxonomy; fixed segmentation windows may underperform in bundle branch block or paced rhythm; prospective clinical evaluation is required before patient-care claims.